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Adsorption Free Energy of a Mobile Physisorbed Molecular Catalyst from Well-Tempered Metadynamics and from the Harmonic Approximation

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Abstract

The adsorption free energy of a molecular catalyst is usually calculated with harmonic frequency analysis of the optimized adsorbed structure, and errors are often blamed on anharmonicity. Here, using fac-Re(bpy)(CO)3Cl on graphene, we show that the main source of error is different. Harmonic analysis treats the complex as vibrating around a single minimum, but at 298 K, the complex can move across the surface, rotate around the surface normal, and rock. In this work, we use multiple-walker well-tempered metadynamics to sample the desorption coordinate and then convert the free-energy profile to a standard-state value, with the gas at 1 bar and the adsorbate at the standard surface concentration as recommended by Campbell et al. The result is ΔG°ads(298 K) = −64.1 ± 2.1 kJ mol−1 at PBE-D3(BJ), while the harmonic approximation gives −26.3 kJ mol−1. Of the 37.8 kJ mol−1 difference, only 1.7 kJ mol−1 comes from the anharmonic shape of the binding well along the surface normal; 36.3 kJ mol−1 is due to the frustrated translations and rotations of the bound complex, which the harmonic model treats as small vibrations around a single minimum. The two exceed the total by the 0.2 kJ mol−1 quantum-to-classical step, which acts in the opposite direction. Insertion-based estimators, which work well for small adsorbates in pores, do not apply here because the ensembles with and without the adsorbate–surface interaction are separated by 34 kBT and do not overlap. The missing entropy cancels out when comparing similar poses of the same adsorbate, but not when referencing the gas phase.

Keywords

adsorption free energy, standard state, frustrated translations, hindered translator, metadynamics, physisorption, molecular catalysts, anharmonicity

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Adsorption Free Energy of a Mobile Physisorbed Molecular Catalyst from Well-Tempered Metadynamics and from the Harmonic Approximation

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ABSTRACT

The adsorption free energy of a molecular catalyst is usually calculated with harmonic frequency analysis of the optimized adsorbed structure, and errors are often blamed on anharmonicity. Here, using *fac*-Re(bpy)$(CO)_3$Cl on graphene, we show that the main source of error is different. Harmonic analysis treats the complex as vibrating around a single minimum, but at 298 K, the complex can move across the surface, rotate around the surface normal, and rock. In this work, we use multiple-walker well-tempered metadynamics to sample the desorption coordinate and then convert the free-energy profile to a standard-state value, with the gas at 1 bar and the adsorbate at the standard surface concentration as recommended by Campbell et al. The result is $\Delta G°_{ads}$(298 K) = −64.1 ± 2.1 kJ mol^{-1} at PBE-D3(BJ), while the harmonic approximation gives −26.3 kJ mol^{-1}. Of the 37.8 kJ mol^{-1} difference, only 1.7 kJ mol^{-1} comes from the anharmonic shape of the binding well along the surface normal; 36.3 kJ mol^{-1} is due to the frustrated translations and rotations of the bound complex, which the harmonic model treats as small vibrations around a single minimum. The two exceed the total by the 0.2 kJ mol^{-1} quantum-to-classical step, which acts in the opposite direction. Insertion-based estimators, which work well for small adsorbates in pores, do not apply here because the ensembles with and without the

adsorbate–surface interaction are separated by 34 $k_B T$ and do not overlap. The missing entropy cancels out when comparing similar poses of the same adsorbate, but not when referencing the gas phase.

KEYWORDS: adsorption free energy, standard state, frustrated translations, hindered translator, metadynamics, physisorption, molecular catalysts, anharmonicity

1. INTRODUCTION

Two methods for calculating the adsorption free energy of *fac*-Re(bpy)$(CO)_3$Cl on graphene at 298 K and 1 bar give very different results. Using the same theoretical level and standard state, the standard rigid-rotor harmonic-oscillator (RRHO) analysis of the optimized adsorbed geometry gives −26.3 kJ mol^{-1}, while well-tempered metadynamics along the surface-normal direction gives −64.1 kJ mol^{-1}. This difference changes the adsorption equilibrium constant by a factor of about 4×10^6, which can determine whether a supported molecular catalyst stays on its support or leaches into solution.

Each value is valid within its respective assumptions, which differ in the permitted motions of the bound state. The harmonic model treats the complex as vibrating around a single minimum. In contrast, the metadynamics simulation enables the complex to slide across the surface, rotate about the surface normal, and rock. During 100 ps of unbiased dynamics at 298 K, the center of mass moves by 2.3 Å root mean square and visits multiple lattice sites. These frustrated translations and rotations are not small vibrations around a single minimum, although the harmonic model treats them as such. As demonstrated below, these motions account for 36.3 of the 37.8 kJ mol^{-1} difference and the anharmonic shape of the potential along the desorption

coordinate contributes 1.7 kJ mol^{-1}, the balance being a 0.2 kJ mol^{-1} quantum-to-classical step of opposite sign.

Molecular catalysts immobilized on carbon via dispersion interactions are widely employed in CO_2 reduction and related electrochemical processes. Examples include pyrene-labeled complexes on graphitic carbon and nanotubes,[1,2] as well as metal complexes retained in supported ionic liquid films.[3,4] The present test system is closely related to a catalyst immobilized in a practical study, specifically Re(*t*Bu-bpy)$(CO)_3$Cl on multi-walled carbon nanotubes.[5] Adsorbed molecular catalysts have also been modeled using static DFT on graphitic carbon[6] and on oxide supports.[7]

The harmonic approximation has already been checked against sampled free energies in the case of small adsorbates in confined pores. Plessow[8] obtained harmonic errors of 10 to 20 kJ mol^{-1} for adsorbates such as CO, propene, and ketene in zeolites, with the magnitude of the errors increasing with the adsorbate's mobility in the pore. A composite insertion procedure has been proposed that achieves the sampled result at an acceptable cost. The cause of these errors is now known: a weakly adsorbed reactant can rotate and translate within the pore, whereas the harmonic model does not permit such movements. Thermodynamic integration[9] and mode-resolved anharmonic treatments[10] also quantify this same effect for adsorbates of that size. In molecular association, where the same harmonic treatment is applied to a host–guest complex rather than to a surface site, dispersion-corrected DFT with meta-GGA or hybrid functionals, a continuum solvation model, and a free-rotor treatment of the low-frequency modes reproduces measured association free energies to about 8 kJ mol^{-1} on average.[11]

No similar test has been done for a supported molecular catalyst. These catalysts are much larger and sit on an open surface instead of inside a pore, so the translational contribution is probably

even bigger. Accurately calculating the adsorption free energy for these systems is especially important for understanding leaching.

Section 3.1 explains why insertion estimators that work for small adsorbates in pores do not apply to larger adsorbates: these methods need overlap between the ensembles with and without the adsorbate–surface interaction, but here, the two are separated by 34 k_BT. Sections 2 and 3 show how we determine the free energy using metadynamics, convert it to a standard-state value with explicit terms, and demonstrate that the difference from the harmonic value comes from the frustrated translations and rotations of the mobile adsorbate.

2. COMPUTATIONAL METHODS

2.1. Protocol

Metadynamics creates a free-energy profile along a selected coordinate by applying a history-dependent bias potential made up of Gaussians (hills) that are placed at the values of that coordinate where the system has previously been, in order to drive the system out of areas it has already visited; the total bias accumulated ends up being the negative of the free-energy profile. In the well-tempered version,12 the height of the hills decreases as the bias increases, thereby ensuring that the profile converges smoothly, and multiple walkers13 follow independent trajectories that both deposit the bias and experience it. Eight walkers shared a single well-tempered bias with respect to the collective variable z, which is defined as the surface-normal distance between the center of mass of the complex and that of the graphene sheet (see Figure 1a). Hills of 1.2 kJ mol−1 were placed every 0.5 ps, with a σz of 0.30 Å and a bias factor of 20. Harmonic walls at z = 2.0 and 14.0 Å were used to keep the walkers within the sampled region and below the periodic image. The calculations were carried out using CP2K14 together with PLUMED,15 using the PBE-D3(BJ)16–18 functional with DZVP-MOLOPT-SR basis sets19 and

a a 400 Ry density cutoff at the Γ point, in a 5 × 5 graphene cell containing 50 carbon atoms (Acell = 131 Å2), with the 28-atom complex adsorbed on top of it. Dynamics were propagated in the canonical ensemble at 298.0 K using canonical sampling through velocity rescaling with a 40 fs time constant, applied to every degree of freedom. The time step was 1 fs, and the masses of the hydrogen atoms were set to 3.0151 amu. Since classical configurational free energies do not depend on mass, this change does not affect the free-energy profile. The total sampling time was 800 ps, which corresponds to 100 ps per walker.

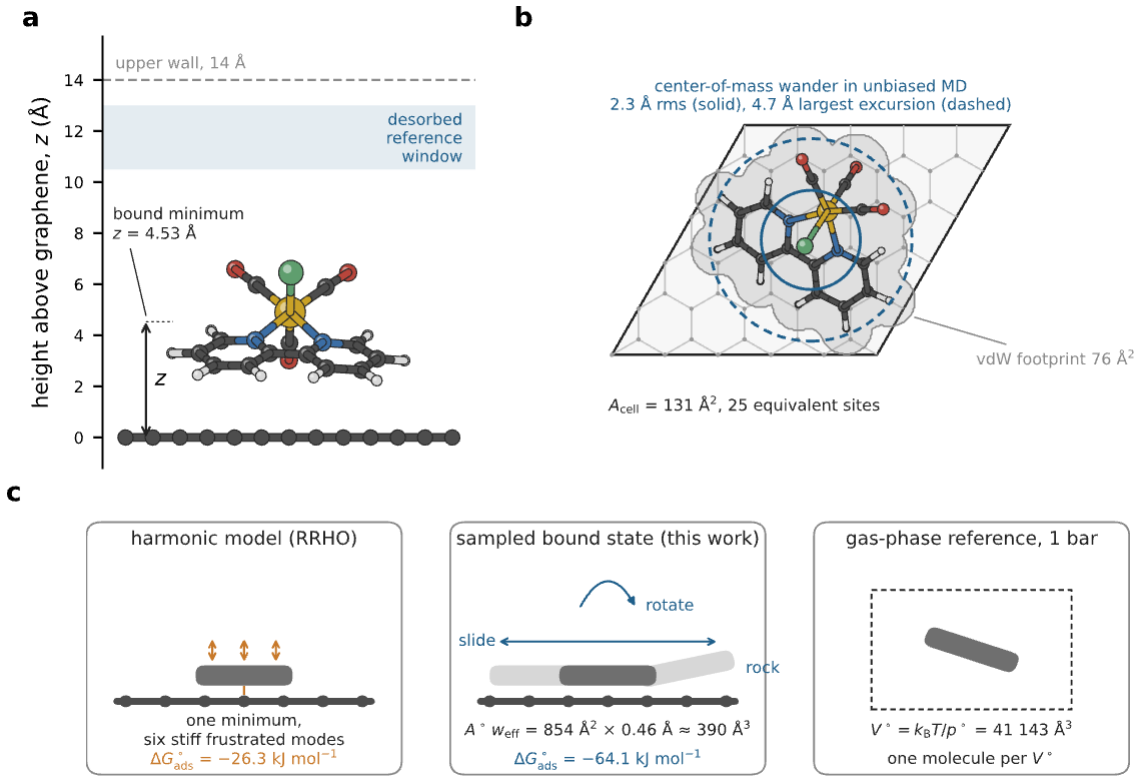


Figure 1. (a) The complex at the bound minimum, 4.53 Å above the graphene sheet, along with the desorbed reference window and the harmonic wall which closes the sampled range. The atom colors are as follows: Re is gold, Cl is green, O is red, N is blue, C is gray, and H is white. (b) The same complex seen from above in the 5 × 5 cell. Circles indicate the in-plane displacement of the center of mass from unbiased dynamics: the root mean square radial spread is 2.3 Å and

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the maximum radial spread is 4.7 Å. The van der Waals footprint, 76 $Å^2$, is the measure of adsorbate size referred to in Section 3.1. (c) The three configurational volumes that are used in the comparison. The harmonic model restricts the complex to one minimum with six stiff modes; the bound state obtained by sampling occupies $A^\circ w_{eff}$ = 854 $Å^2$ × 0.457 Å = 390 $Å^3$ at the standard surface concentration; the gas-phase reference at 1 bar permits one molecule per V° = 41 143 $Å^3$. ΔG°_{ads} is equal to $-k_B T$ times the logarithm of the ratio of the bound to free volume, the well depth appearing via the Boltzmann factor in equation 1.

2.2. Insertion and perturbation estimators

In insertion approaches, molecules are sampled separately, then placed at random positions and orientations above the surface, and the interaction energy ΔU of each trial placement is recorded. The free energy is obtained from the Boltzmann-weighted average over the trials, −kBT ln⟨exp(−ΔU/kBT)⟩. Widom's method20 evaluates this average directly. The Bennett acceptance ratio (BAR)21 also utilizes configurations sampled from the bound state, improving the estimate when the two ensembles partially overlap. Plessow8 introduces a hard-sphere pre-screen to reject placements overlapping surface atoms without requiring an electronic-structure calculation. All variants require that a significant fraction of trial placements resemble the bound state, ensuring overlap in the ΔU distribution. When this is not the case, the average is dominated by a few out of millions of trials, and the statistical error calculated from the remainder significantly underestimates the true uncertainty.22 For this reason, Plessow notes that the method is not expected to work for strongly bound, localized adsorbates.8

2.3. Conversion of F(z) to a standard-state adsorption free energy

The fact that the trajectory is biased means that the result is a free-energy profile $F(z)$ rather than a single adsorption free energy. To obtain a standard-state value from *F(z)*, three choices have to

be made: which range of z is to be regarded as corresponding to the bound state, which range far from the surface is to serve as the reference for desorption, and which standard states are to be used for the gas and for the adsorbate. Each of these choices is given as a row in Table 1. The standard adsorption free energy can then be expressed as:

$$\Delta G°_{ads} = -k_B T \ln(A° \, w_{bind} / V°), \qquad w_{bind} = \int_{bound} \exp\{-[F(z) - F_{plateau}] / k_B T\} \, dz \tag{1}$$

At a temperature of 298 K and a pressure of 1 bar, V° equals $k_B T/p°$, that is, 41 143 Å^3. $A° w_{bind}$ represents the configurational volume available to the bound molecule at the standard surface concentration, where $A° = 1/\Gamma° = 854$ Å^2 is the area per molecule in the adsorbate standard state (as explained below), and $V°$ is the volume per molecule of an ideal gas at a pressure of 1 bar; thus, $\Delta G°_{ads}$ is equal to $-k_B T$ multiplied by the logarithm of the ratio of these two quantities. The integral extends over the bound region and is referenced to the free energy of the desorbed plateau, which is the flat section of $F(z)$ far from the surface where the molecule no longer interacts with the surface. By dividing w_{bind} into the depth of the well and its effective width, $w_{bind} = w_{eff} \exp(D_F/k_B T)$, the two components are separated.

$$\Delta G°_{ads} = -D_F + k_B T \ln[V° / (A° \, w_{eff})] \tag{2}$$

The first term accounts for the potential energy surface, including its anharmonicity in the z direction, while the second term represents the entropy loss when a molecule, which was originally free in a volume $V°$, is confined to the bound-state volume $A° w_{eff}$. Table 1 provides both of these values.

The area $A°$ mentioned in equation 1 is the area per molecule corresponding to the standard state of the adsorbate and it is not a characteristic of the simulation cell. In the dilute limit the equilibrium ratio between the surface concentration of the adsorbate, Γ, and its gas-phase concentration, c, is equal to w_{bind}, and in this ratio the cell area cancels out (Section S2.3). A

standard-state value therefore involves selecting a standard surface concentration $\Gamma^\circ = 1/A^\circ$, and two methods can be compared only if they use the same value.[23] We have adopted the value suggested by Campbell et al., $\Gamma^\circ = e^{1/3}(N_A p^\circ/RT)^{2/3}$, at which a two-dimensional ideal gas has two-thirds of the translational entropy of the three-dimensional ideal gas at 1 bar.[23] At 298 K this results in $\Gamma^\circ = 1.17 \times 10^{13}$ cm^{-2}. The same concentration is applied to the harmonic value (Section 2.5). Values for another standard concentration follow from $\Delta G^\circ_{ads}(\Gamma^{\circ\prime}) = \Delta G^\circ_{ads}(\Gamma^\circ) + k_B T \ln(\Gamma^{\circ\prime}/\Gamma^\circ)$. There is no need for a separate term in equation 1 to account for the in-plane motion of the bound molecule since at every z the trajectory samples both the in-plane position and the orientation without bias, the surface corrugation thus being included in $F(z)$.

Table 1. The reference-state ledger for the standard Gibbs energy of adsorption (298 K, gas at 1 bar, adsorbate at $\Gamma^\circ = 1.17 \times 10^{13}$ cm^{-2}) as derived from the 800 ps surface.

Term	Value	Sensitive to	Varied in
Well depth D_F	75.6 kJ mol^{-1}	bias convergence, plateau window	Section 2.4, S2
Plateau window	10.5–13.0 Å	flatness of $F(z)$	Section 2.4, S2
Bound cutoff	$z \leq 9.0$ Å	nothing at this well depth	Section 2.4
Effective width w_{eff}	0.457 Å	σ_z of the hills	S2
Standard-state area $A^\circ = 1/\Gamma^\circ$	854 Å^2	choice of standard state, T	Section 2.3, S2.3
Reference volume V°	41 143 Å^3	T, p°	—
Confinement term	+11.5 kJ mol^{-1}	the three rows above	—
ΔG°_{ads}	−64.1 ± 2.1 kJ mol^{-1}		Section 3.5

2.4. Convergence and statistical error

The adsorption free energy values obtained from the 600, 700 and 800 ps surfaces are −65.7, −61.6 and −64.1 kJ mol^{-1} (as shown in Figure 2b), with a standard deviation of 2.1 kJ mol^{-1}. The drift during the first half of the run is greater than the scatter and disappears by 600 ps. Altering

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the plateau window from z = 10.0 to 13.5 Å results in a change of at most 0.4 kJ mol^{-1}, and changing the bound-region cutoff from 8 to 10 Å results in a variation of less than 0.01 kJ mol^{-1} because, at a well depth of 30 k_BT, the integrand in equation 1 is negligible at the cutoff. More details on the per-walker diagnostics and the sensitivity scan can be found in Section S2.

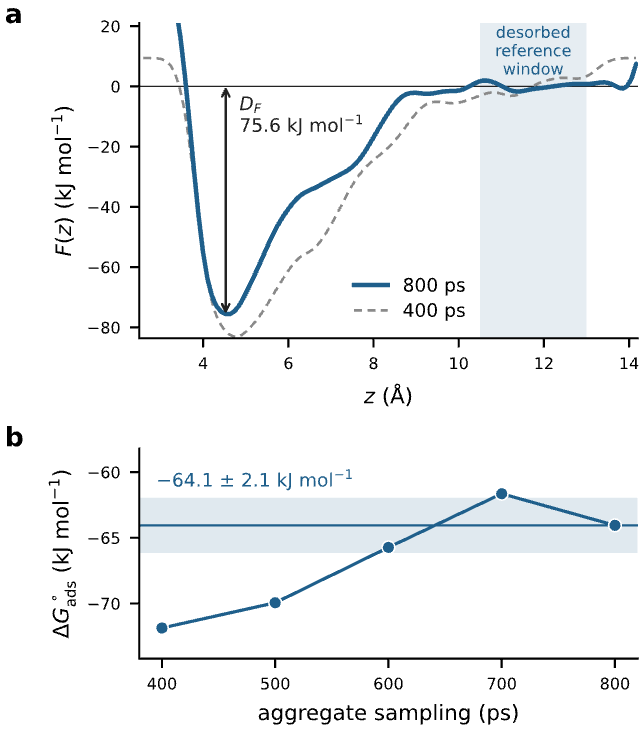


Figure 2. (a) The free-energy surface along *the z* direction for both the 800 ps aggregate sampling (solid line) and the 400 ps sampling (dashed line), each referenced to its respective desorbed plateau. The plateau window denotes the interval over which the desorbed reference value is averaged, and D_F represents the well depth as defined in equation 2. Panel (b) presents the cumulative estimates of ΔG°_{ads}, with the shaded band indicating the ±2.1 kJ mol^{-1} standard deviation of the final three estimates.

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2.5. The harmonic reference

The harmonic reference was evaluated in the normal manner by taking the partial Hessian of the 28-atom complex at its optimized adsorbed geometry, considering all 3N = 84 eigenvalues as vibrational modes, and comparing it with that of the free molecule which has 3N − 6 = 78 vibrational modes, together with translation in an ideal gas and a rigid rotor having σ = 1 at a pressure of 1 bar. Zero-point energies have been included on both sides and differ by 2.1 kJ mol−1. The harmonic value relates to one minimum; in the lattice-gas description this corresponds to a standard coverage $\theta^\circ = 1/2$ of that minimum, at which the coverage term $k_B T \ln[\theta^\circ/(1 - \theta^\circ)]$ is zero. For the standard state described in Section 2.3, the coverage of the minima is $\theta^\circ = A_{site}\Gamma^\circ = 6.1 \times 10^{-3}$, where $A_{site} = 5.24$ Å^2 is the area of the graphene primitive cell, this cell containing one instance of each minimum. The coverage term is then −12.6 kJ mol^{-1} and the harmonic value thus amounts to −26.3 kJ mol^{-1}.[23] The partial Hessian keeps the graphene sheet fixed; when the analysis is repeated using the full 78-atom Hessian and correcting for the finite mass of the periodic sheet, the harmonic value is altered by −0.6 kJ mol^{-1} (Section S6.2).

3. RESULTS AND DISCUSSION

3.1. Why insertion estimators fail for a catalyst-sized adsorbate

Before employing the metadynamics approach, the insertion route was tested; however, this calculation produced no usable free energy because the insertion weight was indistinguishable from zero. For CO in a zeolite, the same method remains more reliable and less computationally demanding than a metadynamics-based approach.[8] The determining factor for the success or failure of insertion approaches is the size of the adsorbate.

Every insertion variant requires overlap between the ΔU distribution of the random trial placements and that of the bound state, that is, a significant fraction of trial placements must

resemble the adsorbed complex. For the present system that fraction is effectively zero, so the estimator yields nothing usable. The static adsorption energy is −84.2 kJ mol^{-1} when using PBE-D3(BJ), meaning that the interaction energies characteristic of the bound ensemble are 34 $k_{\mathrm{B}}T$ lower than those typical of the ensemble in the absence of interaction. The question of whether the two ensembles overlap is decided by comparing the separation with the widths of the two ΔU distributions. Since ΔU is the sum of many approximately independent atom–surface contacts, just as is true of any such sum, the mean increases with the number of contacts N while the thermal spread increases only in proportion to the square root of N. In the case of the twenty-or-so atoms in the complex that are in van der Waals contact with the graphene sheet, the spread is on the order of a few $k_{\mathrm{B}}T$, hence the two distributions are separated by about ten times their width. The separation increases with adsorbate size, since N does too, and the van der Waals footprint of the complex, as projected onto the surface plane (76 Å^2), compared with 12 Å^2 for CO by the same method, is the practical measure of N. For comparison, the interaction terms in Plessow's zeolite study[8] are 18 to 44 kJ mol^{-1} at temperatures between 473 and 673 K, or 3 to 11 $k_{\mathrm{B}}T$, and even in that case overlap had to be assisted by a hard-sphere intermediate.

3.2. Comparison with the harmonic approximation

The value assigned by RRHO to one minimum is −13.7 kJ mol^{-1}, or −26.3 kJ mol^{-1} on the standard state of Section 2.3; the calculation itself is straightforward; the difficulty lies in identifying which bound state it corresponds to.

The harmonic analysis of the adsorbed complex replaces the three translations and three rotations of the free molecule with six frustrated modes at a single minimum; a frustrated mode is one in which a translation or rotation becomes a low-frequency vibration when the molecule is adsorbed. By means of this analysis the entire translational and rotational free energy present in

the gas phase (equal to +87.8 kJ mol^{-1} at 1 bar) is eliminated, and the change in vibrational free energy on adsorption (−17.3 kJ mol^{-1}, taken over all 84 modes of the complex against the 78 internal modes of the free molecule) is added in its place; the two contributions give a harmonic thermal term of +70.5 kJ mol^{-1} for one minimum and +57.9 kJ mol^{-1} on the standard state of Section 2.3 (Table S5). The metadynamics approach substitutes these six degrees of freedom by confining the molecule to the volume $A°w_{eff}$ and by taking into account the well depth in free energy with respect to the static adsorption energy; together, these two factors yield a sampled thermal term of +20.1 kJ mol^{-1} (Table S5). The difference between the two thermal terms is 37.8 kJ mol^{-1}.

3.3. Decomposition of the difference between the two methods

Figure 3 shows the progression from the harmonic approximation to the metadynamics-sampled value in three steps that link four levels, two of which are small. The harmonic value for one minimum is first placed on the standard state of Section 2.3 (−12.6 kJ mol^{-1}, Section 2.5). In the first step, both methods are brought into agreement by acknowledging that the molecular dynamics are classical, do not include zero-point energy, and that when the harmonic partition functions are evaluated classically a shift of +0.2 kJ mol^{-1} is obtained. The second step involves replacing the harmonic description of the motion along *the z-axis* with the sampled result, which gives a value of −1.7 kJ mol^{-1}. The thermal width of the well is 0.326 Å in the unbiased dynamics as opposed to 0.165 Å in the harmonic model; the k_BT ln of the ratio of these values gives the complete anharmonic correction along the desorption coordinate and accounts for 4.4% of the difference. The third step, −36.3 kJ mol^{-1}, is the contribution of the frustrated translations and rotations (hereafter the frustrated modes): the entropic contribution from the five remaining degrees of freedom, all of which the harmonic model treats as small vibrations around one

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minimum—namely, two in-plane translations, rotation about the surface normal, and two rocking motions.

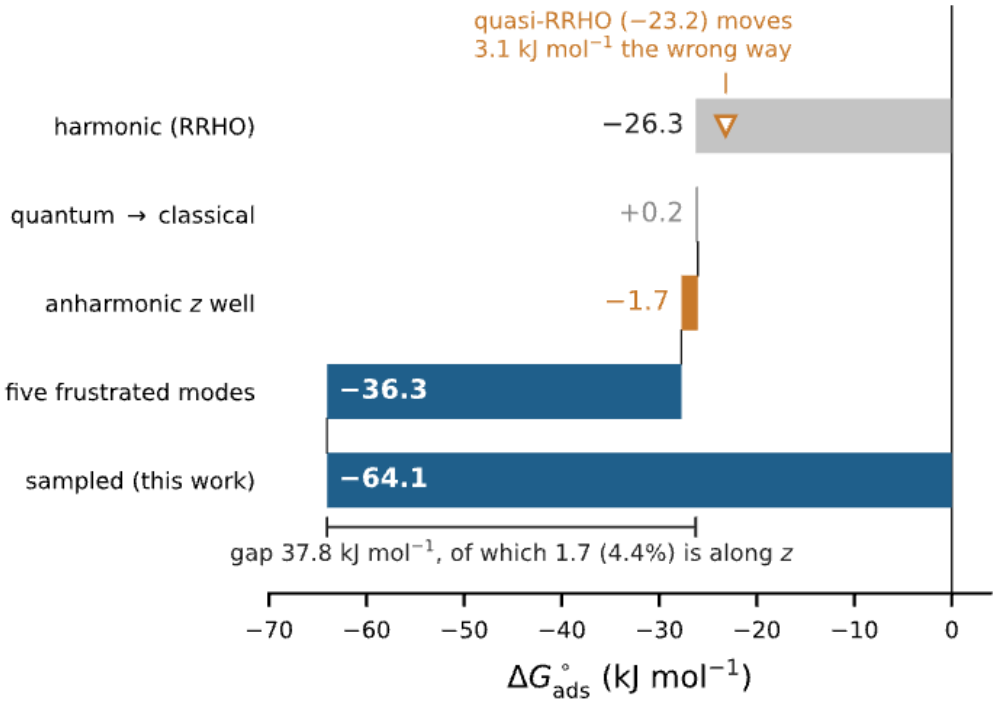


Figure 3. Decomposition of the difference between the harmonic approximation and the value obtained from metadynamics sampling, on the standard state of Section 2.3. There are three stages linking four levels: the RRHO result for one minimum on that standard state (−26.3 kJ mol^{-1}, or −13.7 kJ mol^{-1} before conversion); the same result but with the harmonic partition functions evaluated classically (an increase of +0.2 kJ mol^{-1}); that result further modified by replacing the harmonic treatment of motion along z with the sampled free-energy profile (a decrease of −1.7 kJ mol^{-1}); and finally the sampled value, which is obtained by adding the frustrated translations and rotations of the five other degrees of freedom (−36.3 kJ mol^{-1}). The orange portion is the only contribution due to the shape of the desorption potential, i.e. the deviation of *F(z)* from a parabolic form along z; the blue portion results from the surface area and orientations accessible to the molecule, which the harmonic model treats as small vibrations

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around one minimum. The open triangle indicates the quasi-RRHO approximation, in which the entropy of the modes below a 50 cm^{-1} cutoff is interpolated to approach free-rotor behavior.

The two-dimensional run (Section S4.2) allows the in-plane component of the frustrated-mode term to be measured directly. When the in-plane position of the center of mass is folded onto the graphene primitive cell, the free-energy landscape for in-plane motion within a single adsorption site is obtained. This represents the sampled version of the translational energy landscapes that Bajpai et al. and Waitt et al. derive from their static DFT scans.[24,25] The corrugation within a site amounts to 2.5–3.3 $k_B T$. The center of mass samples 0.8–2.1 $Å^2$ of each 5.24 $Å^2$ cell, as opposed to 0.039 $Å^2$ in the classical harmonic model, and this results in an anharmonicity of the two frustrated translations of 5–10 kJ mol^{-1}. The remaining 27–31 kJ mol^{-1} relates to rotation about the surface normal and the two rocking motions; it includes $k_B T \ln 6$ = 4.4 kJ mol^{-1} for the six orientations that are equivalent under the six-fold axis of graphene, a contribution that a single-minimum harmonic calculation fails to take into account.[26] Since the barriers between adjacent positions are a few $k_B T$, the bound state is not made up of a number of separate harmonic minima and therefore counting the minima does not yield the sampled value. This is the situation described by the hindered-translator and hindered-rotor models.[26]

The quasi-RRHO method[11] for dealing with soft modes does not offer a solution. This approach aims to overcome the shortcomings of RRHO by replacing all “soft modes” (those below 50 cm^{-1}) with the entropic contribution of a free rotor, on the basis that these “soft modes” correspond to the two-dimensional translation, rotation about the surface normal, and rocking movements which are not properly sampled by RRHO. Using a cutoff at 50 cm^{-1} gives a value of −23.2 kJ mol^{-1} on the standard state of Section 2.3 (−10.6 kJ mol^{-1} for one minimum), which is 3.1 kJ mol^{-1} further from the sampled value than the simple RRHO result (Figure 3). Although

the direction of this correction might depend on the system, it is impossible to close the remaining difference (the "soft modes" would have to have frequencies of 5 cm^{-1}, which is unrealistic for a bound system). The hindered-translator and hindered-rotor models[26] deal with the same frustrated degrees of freedom explicitly, interpolating between the harmonic limit and free translation or free rotation, but they take the corrugation and rotation barriers of the bound state as their input rather than a frequency cutoff.

3.4. Dependence on adsorbate size and mobility

The error magnitude in the harmonic approximation depends on the mobility of the adsorbate, more precisely on the degree of translation and rotation that is allowed in the bound state, not on the softness of the internal modes. Earlier investigations of small adsorbates in pores[8] found harmonic errors ranging from 10 to 20 kJ mol^{-1} since the pore wall prevents in-plane translation irrespective of how the model is treated. On an open surface, however, this kind of translation is not restricted, and for adsorbates of catalyst size the error rises to 37.8 kJ mol^{-1}.

A lesser contribution to the error is due to the mass of the adsorbate. For a molecule which is free to move over an area A, the translational free energy has the term $-k_B T \ln(A/\Lambda^2)$, where Λ is the thermal de Broglie wavelength. The size of this term increases with the mass of the adsorbate in proportion to $k_B T \ln m$. When going from CO to this complex, this factor alone adds 6.9 kJ mol^{-1}, regardless of the stiffness of the frustrated modes or the available area.

3.5. Sources of error

The sources of error are as follows:

Statistical, ±2.1 kJ mol^{-1}. The spread of the three cumulative block estimates in the last 200 ps.

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Reference-window definition, ±0.4 kJ mol^{-1}. Plateau-window and bound-cutoff scans, Section 2.4.

3.6. Method selection

Figure 4 sets out the various methodological approaches available for accurately calculating the adsorption free energy of a molecular catalyst.

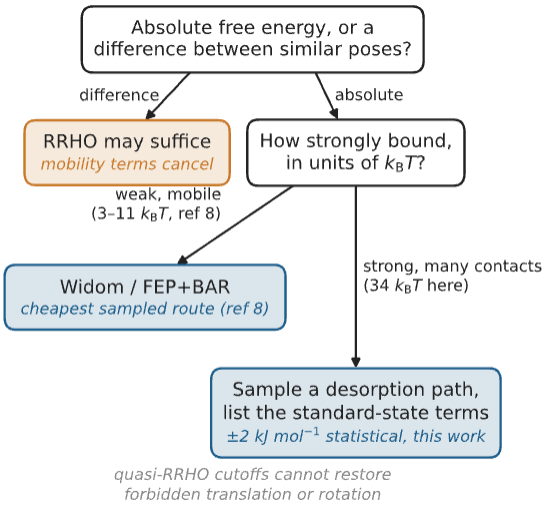


Figure 4. Three methodological options: harmonic analysis, the insertion estimator, or the sampled desorption path. In the cases having interaction terms of 3 to 11 k_BT, the insertion estimators do converge, whereas in the present case with an interaction strength of 34 k_BT, convergence is not obtained. The exact boundary between these two regimes has not yet been established. The errors expected correspond to those given in this paper and in reference 8.

For adsorbates that are weakly bound and mobile, such as small molecules within zeolite pores with interaction energies between 3 and 11 k_BT, the insertion method is the most cost-effective sampling approach. In contrast, when the adsorbate is strongly bound due to multiple surface interactions (as in the present case at 34 k_BT), the two ensembles do not overlap (see

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Section 3.1), and the only viable strategy is to perform sampling along a desorption coordinate with explicit standard-state conversion.

4. CONCLUSIONS

With respect to *fac*-Re(bpy)(CO)3Cl on graphene, harmonic frequency analysis fails to account for 37.8 kJ mol^{-1} of the sampled adsorption free energy of −64.1 ± 2.1 kJ mol^{-1} (at 298 K and 1 bar, with the adsorbate at the standard surface concentration of Campbell et al.). Only 1.7 kJ mol^{-1} of this difference is due to the anharmonic shape of the binding well along the surface normal; the other 36.3 kJ mol^{-1} results from the frustrated translations and rotations, which the harmonic approach treats as small vibrations around one minimum. The two in-plane translations account for 5 to 10 kJ mol^{-1} since the corrugation within each adsorption site is only a few $k_B T$ and the molecule occupies a much larger area of each site than the harmonic model permits. The rotation about the surface normal and the rocking contribute the remaining 27 to 31 kJ mol^{-1}. A frequency cutoff of the quasi-RRHO type fails to recover these contributions (Section 3.3).

In this approach, a desorption coordinate is selected, and the resulting profile is transformed into a standard state using an explicit reference-state ledger (Table 1). The bound range, desorbed reference, and standard states of the gas and the adsorbate are all parameters with defined sensitivities, allowing readers to adjust them rather than relying on implicit values from a frequency calculation.

Harmonic analysis is suitable in those cases where the mobility terms cancel out, such as when making relative comparisons between different sites, poses or conformers of the same adsorbate on the same surface; the absolute adsorption free energy, however, when referenced to the gas phase or a solution, includes the full error.

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Whether a supported molecular catalyst stays on the support depends on this absolute value. It is essential to calculate the adsorption free energy accurately when modeling physisorbed molecular catalysts, because features like leaching and boomerang behavior, where a detached complex re-adsorbs later in the catalytic cycle, depend on having accurate free energies.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.XXXXXXX.

Input files; free-energy surface construction and sensitivity scans; the standard state of the adsorbate; the decomposition of the difference between the two methods; the two-dimensional dataset and the in-plane free-energy landscape within one adsorption site; insertion diagnostics; the harmonic reference, its variants and the full-Hessian test; reproducibility and data availability (PDF)

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Notes

The author declares no competing financial interest.

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DATA AVAILABILITY

The reduced one-dimensional dataset, the full two-dimensional hills and COLVAR files, the vibrational analysis of the adsorbed complex and its inputs, the analysis scripts, and numbers.yaml are deposited at Zenodo, DOI: 10.5281/zenodo.22301274.

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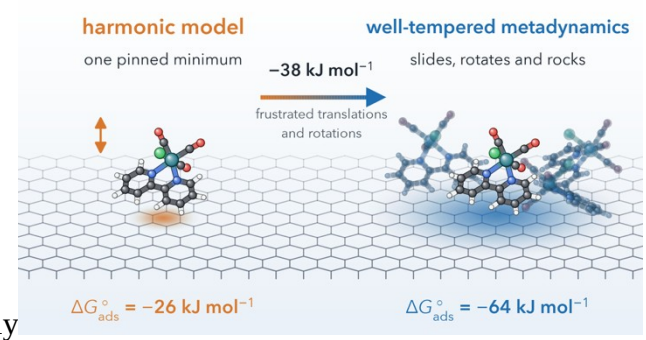
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